MIL-S-10446B(EL)
13 October 1970

Superseding
MIL-S-10446A(SigC)
18 February 1957

MILITARY SPECIFICATION

SWITCHBOARD, TELEPHONE, MANUAL SB-86( )/P

1. SCOPE

1.1 This specification covers one type of local battery telephone switchboard designated as: Switchboard, Telephone, Manual SB-86( )/P (See 6.1 and 6.3) consisting of:

Switchboard Signal Assembly TA-207( )/P (jack section)
Switchboard Section, Telephone, Manual SB-248( )/P (keyshelf section) containing:
  Telephone Circuit, Cord TA-208( )/P (cord pack)
  Telephone Circuit, Operator's TA-220( )/P (operator's pack)
Power Supply PP-990( )/G (power pack)
Headset-Microphone H-91( )/U (operator's telephone)
Cover, Switchboard, Telephone, Manual CY-3725( )/P (transit case)

2. APPLICABLE DOCUMENT

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strapping, Flat
PPP-B-601	Boxes; Wood, Cleated-Plywood
PPP-B-621	Boxes; Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container grade), and Cut Shapes
PPP-T-76	Tape; Pressure-Sensitive Adhesive Paper, (For Carton Sealing)

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MIL-B-18/34	Battery, Dry BA-200/U
MIL-P-116	Preservation, Methods of.
MIL-B-117	Bags, Interior Packaging
MIL-H-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-H-12576	Headset-micro. H-91( )/U, H-92( )/U,H-144H/U
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-188	Military Communication System Technical Standards
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-135947	Power Supply PP-990( )/G
SC-DL-136100	Cover, Switchboard, Telephone, Manual CY-3725( )/P
SC-DL-136149	Switchboard, Telephone, Manual SB-86( )/P
SC-DL-136200	Telephone Circuit, Cord TA-208( )/P
SC-DL-136282	Telephone Circuit, Operator's TA-220( )/P
SC-DL-136400	Switchboard Signal Assembly TA-207( )/P
SC-DL-136600	Switchboard Section, Telephone, Manual SB-248( )/P

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions may be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First Article Samples.- The contractor shall furnish two (2) First Article samples for approval, if required by the invitation for bids and contract (See 4.3).

3.2 Construction.- The Switchboard, Telephone, Manual SB-86( )/P shall be constructed in accordance with the requirements of this specification and the drawings listed on Drawing and Data List SC-DL-136149. The following units of the switchboard shall be constructed in accordance with the indicated Drawing and Data List (See 4.4):

Power Supply PP-990( )/G	SC-DL-135947
Cover, Switchboard, Telephone, Manual CY-3725( )/P	SC-DL-136100
Telephone Circuit, Cord TA-208( )/P	SC-DL-136200
Telephone Circuit, Operator's TA-220( )/P	SC-DL-136282
Switchboard Signal Assembly TA-207( )/P	SC-DL-136400
Switchboard Section, Telephone, Manual SB-248( )/P	SC-DL-136600

3.3 Parts, Materials, and Processes; General.- In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply (See 4.4).

3.4 Finish.- The equipment shall be finished in accordance with MIL-F-14072 and the equipment drawings (See 4.4).

3.5 Marking.-

3.5.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4) Panel markings shall be Group I as described in that specification.

3.5.2 Visibility.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial Numbers.- The following units shall be serial numbered:

Switchboard, Telephone, Manual SB-86( )/P
Switchboard Section, Telephone, Manual SB-248( )/P
Switchboard, Signal Assembly TA-207( )/P
Power Supply PP-990( )/G

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3.6 Electrical.-

3.6.1 System.- The following system requirements shall be met with the line selector switch in any of its three positions (M, C or T) (See 4.7.1).

3.6.1.1 Transmission loss.- The transmission (insertion) loss between any pair of line binding posts to another pair of line binding posts connected by a cord pack shall not exceed 1.5 decibels (db) for any frequency between 300 and 3500 Hertz (Hz) (See 4.7.1.1).

3.6.1.2 Crosstalk.- The ratio of the induced power in any circuit (disturbed circuit) to the power in the disturbing circuit shall be less than -60 db (See 4.7.1.2).

3.6.1.3 Envelope delay distortion.- The maximum difference of the envelope delay characteristic between pairs of line binding posts connected by a cord pack, in the frequency band between 600 to 3500 Hz, shall not exceed 150 microseconds (See 4.7.1.3).

3.6.1.4 Bridging loss.- The bridging (insertion) loss between any pair of line binding posts to another pair of line binding posts connected by a cord pack and the operator's pack shall be in accordance with the following (See 4.7.1.4):

Frequency (Hz)	Loss (db max.)
300	4.4
600	2.0
1000	1.5
2000	1.5
3000	1.5
3500	1.5

3.6.1.5 Noise.- The noise level of any transmission circuit shall not exceed 18 decibels above reference noise (dbrn) (0 dbrn equals one micro micro watt) (See 4.7.1.5).

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3.6.1.6 Operation.- The equipment shall meet the operational tests of 4.7.1.6 for the following conditions:

a. Magneto line: 0 to 5000 ohm loop, Infinite to 10000 ohm leakage, 15 to 26.5 volts direct current (vdc).

b. Common battery signalling line: 0 to 1000 ohm loop, Infinite to 10000 ohm leakage, 20 to 26.5 vdc.

c. Common battery signalling trunk: 0 to 2000 ohm loop, Infinite to 25000 ohm leakage, 20 to 26.5 vdc.

3.6.1.6.1 Civilian trunk.- Lines 29 and 30 of the switchboard shall be operable as trunk lines with a civilian common battery telephone exchange.

3.6.1.6.2 Conference calls.- Conference circuits shall be established by connecting switchboard cord circuits.

3.6.2 Power Supply PP-990( )/P (See 4.7.2).- With ten each Batteries BA-200( )/U conforming to MIL-B-18/34 installed, the equipment shall provide the following:

3.6.2.1 Metering.- The Direct Current (DC) output voltage to the switchboard and the DC output of the"B Set" of batteries shall be measured with an accuracy of ± 2% of full scale reading by the panel mounted voltmeter.

3.6.2.2 DC output voltage.- The equipment shall deliver to the 24 volt output terminals the voltage from either:

a. Four of five batteries designated as the "A Set" (LOW position of the INT SWBD BATT switch),

b. Five of five batteries designated as the "A Set" (HI position of the INT SWBD BATT switch),

c. The input terminals marked EXT BATT 24V, or

d. The parallel combination of the A and B set of batteries when the shunt terminal is connected to the negative External Battery terminal.

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3.6.2.3 AC output voltage.- With 20 to 32 vdc applied to the input of the vibrator circuit, the Ringing Supply output voltage shall:

a. Cause the Indicator Lamp to light.

b. Have a stable frequency of 20 ± 2 Hz.

c. Be approximately a square wave.

d. Reach 95% of its peak within 250 milliseconds after application of the input voltage.

e. Meet the following requirements:

Input vdc	Input cur. (ma)	Output volts (peak)	Load resistance
24	70 max	90 min	open circuit
24	180 max	80 min	2000 ohms
24	300 max	70 min	1000 ohms
20	N/A	Square wave	2000 ohms
32	N/A	200 max	open circuit

3.6.3 Insulation Resistance.- The insulation resistance between any operating circuit and framework, ground or adjacent operating circuit shall be 100 megohms minimum (See 4.7.3).

3.7 Service conditions.- The equipment shall meet the following service conditions:

3.7.1 Temperature.-

3.7.1.1 Low.- During and after testing as specified in 4.8.1.1, the equipment shall meet the requirements of 3.6.1.6 and 3.6.2.3.

3.7.1.2 High.- During and after testing as specified in 4.8.1.2, the equipment shall meet the requirements of 3.6.1.6 and 3.6.2.3.

3.7.2 Humidity (Moisture-Resistance).- During and after testing as specified in 4.8.2, the equipment shall meet the requirements of 3.6.1.1, 3.6.1.2, 3.6.2.3 and 3.6.3.

3.7.3 Altitude (Elevation).- The equipment shall meet full specification performance for 3.6.1.6 and 3.6.2.3 both during and after testing as specified in 4.8.3.

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3.7.4 Leakage (immersion).- The equipment shall show no evidence of leakage after it is tested as specified in 4.8.4. This requirement applies to the PP-990( )/G and to the SB-248( )/P in the cover (CY-3725( )/P).

3.7.5 Fungus.- The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.8.5. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.7.6 Vibration and bounce.-

3.7.6.1 Vibration, internal.- The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hz (See 4.8.6.1).

3.7.6.2 Bounce.- The equipment shall meet specified performance after subjection to the test of 4.8.6.2. No physical damage except minor surface abrasion shall result from this test.

3.7.7 Shock, Bench handling.- The equipment shall meet specified performance after subjection to the test of 4.8.7.

3.8 Interchangeability.- Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454(See 4.9).

3.9 Electromagnetic interference characteristics.- The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461 for RE02 and A-CE1. Tests and test methods shall be as specified in MIL-STD-462(See 4.4).

3.10 Preconditioning.-

3.10.1 Bounce.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.10.2 Electrical.- The Power Supply PP-990( )/G shall be capable of meeting the requirements herein, without subsequent processing, after 24 consecutive hours of failure-free operation (ringing) (See 4.6.2).

3.11 Air Seal.- The air seal valve in Cover, Swbd, Tel, Manual CY-3725( )/P shall maintain a 1 pound per square inch gage (psig) vacuum inside the sealed cover for one minute (See 4.10).

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3.12 Headset-microphone H-91( )/U.- Headset-microphone H-91( )/U shall conform to MIL-H-12576 (See 4.4).

3.13 Technical literature, tools, and running spare parts.- Technical literature, tools, and running spare parts shall be furnished, if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (See 6.2).

3.14 Workmanship.- The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs (See 4.11):

a. In MIL-P-11268:

General requirements for plastic material and plastic parts.
Wiring and cabling, including:
Slack.
Protection.
Clearance.
Splicing.
Connections, general.
Grounding, general.
Shielding on wire and cable.
Soldering and brazing.
Cleaning of units.
General requirements for securing of parts.
Process for welding.
Cleaning after welding.
Controls.
Self-tapping screws.
Cleaning prior to welding.

b. In this specification: Paragraph 3.2 Construction.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

a. First article inspection (does not include preparation for delivery) (See 4.3).

b. Inspection covered by subsidiary documents (See 4.4).

c. Quality conformance inspection.

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4.2 Classification of inspection (Contd).-

(1) Quality conformance inspection of equipment before preparation before delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

4.3 First Article inspection.- This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B and Group D (see Tables III, IV and V). The First Article (and Group D) inspection shall be performed in the order shown in Table II. Half the samples shall be subjected to the Group I tests and half to the Group II tests. (An odd sample in Group D inspection shall be tested in Group I).

Table II - Order of First Article (and Group D) Inspection

Inspection (Note 1)	Group I	Group II
Non-damaging		
Low Temperature	1	
High Temperature	2	
Elevation		1
Immersion		2
Vibration		3
Potentially damaging		
Moisture resistance	3	
Bounce		4
Shock, bench handling		5
Fungus (See Note 2)	4	

NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the non-damaging tests.

NOTE 2: The Fungus Test shall be performed on First Articles only, or if no First Articles, on a sample unit from Group D inspection. The unit subjected to test shall not be delivered to the Government, it shall be in addition to the quantities to be delivered on contract.

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where Required
Construction	3.2
Parts, materials and processes	3.3
Finish	3.4
Marking	3.5
Electromagnetic interference	3.9
Headset-Microphone H-91( )/U	3.12
Literature, tools and spares	3.13

4.5 Quality conformance inspection of equipment before preparation for delivery.- Each unit of each lot of equipment shall be subjected to preconditioning (see 3.10) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.4). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection level III and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the airseal (4.10) shall be next to last and the operational inspection (4.7.1.6) shall be last prior to packaging.

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Table III - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.14	4.11		
Switchboard Signal Assembly TA-207( )/P			1.0%	4.0%
Switchboard Section, Telephone, Manual SB-248( )/P			1.0%	4.0%
Power Supply PP-990( )/G			1.0%	4.0%
Cover, Swbd, Tel, Man CY-3725( )/P			1.0%	4.0%
Electrical				*
Transmission Loss	3.6.1.1	4.7.1.1	1.0%	
Operation	3.6.1.6	4.7.1.6	1.0%	
Power Supply PP-990( )/P	3.6.2	4.7.2	1.0%	

* All electrical defects are considered major.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table IV and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans shall be Inspection Level S-4 and an AQL for each characteristic test of 4%.

4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the Government.

Table IV - Group B Inspection

Inspection	Req Para	Insp Para
Crosstalk	3.6.1.2	4.7.1.2
Envelope Delay Distortion	3.6.1.3	4.7.1.3
Bridging Loss	3.6.1.4	4.7.1.4
Noise	3.6.1.5	4.7.1.5
Insulation Resistance	3.6.3	4.7.3
Interchangeability	3.8	4.9
Air Seal	3.11	4.10

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4.5.3 Group C inspection.- Not required.

4.5.4 Group D inspection.- This inspection, including sampling, shall conform to Table V and to special procedures for small sample inspection of MIL-STD-105. Group D inspection shall normally be performed on inspection lots that have passed Group A and B inspection and on samples selected from units that have been subjected to and met the Group A and B inspection. At the contractor's option, a Group D lot may consist of up to three Group A and B lots.

Table V - Group D Inspection

Inspection	Req. Para.	Insp. Para.
Temperature	3.7.1	4.8.1
Humidity	3.7.2	4.8.2
Altitude	3.7.3	4.8.3
Leakage	3.7.4	4.8.4
* Fungus	3.7.5	4.8.5
Vibration	3.7.6.1	4.8.6.1
Bounce	3.7.6.2	4.8.6.2
Shock	3.7.7	4.8.7

* Performed on one of the First Articles or units from first Group D sample.

4.5.4.1 Sampling.- The Group D sampling plan shall conform to inspection level S-1 and a 4% AQL for the group.

4.5.5 Disposition of nonconforming product.- Disposition of rejected lots and defective sample units shall be in accordance with MIL-STD-105. When resubmitted for acceptance, such product shall be suitably tagged or identified to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is resubmitted and shall become the property of the Government.

4.5.6 Reinspection of conforming Group D sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract provided all damage is repaired and the units are resubjected to and pass Group A inspection.

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4.6 Preconditioning.- The preconditioning shall precede all other electrical tests of the equipment. Bounce preconditioning shall precede electrical preconditioning of the Power Supply PP-990( )/G.

4.6.1 Bounce.- The equipment shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit of each lot of equipment shall be subjected to this bounce preconditioning for 1 minute prior to selection of units for inspection. After bounce preconditioning the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.10.1)

4.6.2 Electrical (Power Supply PP-990( )/G only).- With an external source of 24 vdc and a 2000 ohm resistor across the Ringing Supply terminals, the ST VIB shall be closed and opened at a cyclic rate of 3 seconds on and ten seconds off. The Ringing Supply shall be operated at this duty cycle for a period of twenty four hours. In the event of failure, and after all repairs are made, the equipment shall be re-subjected to the 24 hour period prior to reinspection.

4.7 Electrical tests.- Unless otherwise specified, electrical tests shall be performed at prevailing room ambient conditions, with 24 vdc battery voltage and at each of the three line selector switch positions (C, M and T). Headset-Microphone H-91( )/U shall be used at the operator's pack except when the operator's pack is used in quantitative measurements in which case the transmitter will be replaced by a 22 ohm ± 1% carbon resistor and the receiver by a 270 ohm ± 1% carbon. resistor.

4.7.1 System.- The following tests shall be performed on Switchboard Signal Assembly TA-207( )/P (jack section) and Manual Telephone Switchboard Section SB-248( )/P (Keyshelf Section).

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by analysis.

4.7.1.1 Transmission loss.- Connect an audio signal generator to a source impedance of 600 ohms and then to the line terminals of the circuit under test. Connect the answer cord of the cord pack to the jack of the input circuit and the call cord to the jack of another line circuit. Terminate the second line circuit in 600 ohms. With the signal generator adjusted for an output voltage of Odbm (one milliwatt in 600 ohms) at the frequencies of 300, 1000, 2000, 3000 and 3500 Hz, measure the voltage across the load resistor. The transmission loss, defined as 20 log of the ratio of the input to the output voltage, shall meet the requirement of 3.6.1.1. Repeat this test for each combination of line circuits and cord packs.

4.7.1.2 Crosstalk.- Connect an operating circuit as in 4.7.1.1 above. Terminate an adjacent circuit at one pair of line terminals with a 600 ohm resistor and at the other pair of line terminals with a Daven Noise Measuring Set #34C, or a Transmission Measuring Set TS-559( )/FT, or equal. At disturbing frequencies of 300, 1000, 2000, 3000 and 3500 Hz and with a generator voltage level of Odbm (one milliwatt in 600 ohm), measure the induced voltage (F1A weighting) in the disturbed circuit. The crosstalk, defined as 20 log of the ratio of the voltage in the disturbed circuit to the voltage in the disturbing circuit shall meet the requirement of 3.6.1.2.

4.7.1.3 Envelope Delay Distortion.- Envelope delay distortion shall be as defined in Appendix B of MIL-STD-188. Connect an operating circuit as in 4.7.1.1 above. Measure the phase delay at frequencies of 600, 700, 1000, 1100, 1500, 1600, 2000, 2100, 2500, 2600, 3000, 3100, 3200, 3300, 3500 and 3600 Hz using a Phase Meter, Ad-Yu Vectorlyzer, or equal. Compute the derivative of Phase Shift versus Frequency curve obtained above and determine the maximum difference of the envelope delay characteristic and compare it with the requirement of 3.6.1.3, Sierra Envelope Delay Test Set Model 340 B, or equal, may be used for this test.

4.7.1.4 Bridging loss.- Connect an operating circuit as in 4.7.1.1 except that the receiver terminals of the operator's pack shall be terminated in 270 ohms ± 1% resistive (and the transmitter contacts left open). Operate the TALK-List switch of the cord pack to TALK and take the Insertion loss measurements at the frequencies cited. The requirements of 3.6.1.4 shall be met.

4.7.1.5 Noise.- Connect an operating circuit as in 4.7.1.1 except that the signal generator shall be replaced with a Daven Noise Measuring Set 34C or equal. Using F1A weighting, the noise level shall not exceed the level stated in 3.6.1.5.

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4.7.1.6 Operational test.- Establish the following circuits and perform the tests as stated:

a. Connect two Telephone Set TA-312( )/PT, or equal to Line binding posts designated as "calling" and "called".

b. Line selector switch on switchboard to be set for respective applications as follows:

M-Magneto
C-Common Battery Signaling
T-Common Battery Signaling Trunk

c. When inserting a plug into a line jack, the associated signal shall go black (or remain black).

d. Both plugs must be out of the cord wells before the supervisory signals operate.

e. All chassis shall be common to the earth ground binding posts.

f. Panel lamps shall light when LAMPS switch is operated to "ON".

4.7.1.6.1 Magneto line test:

a. Limits:

0 to 5000 ohms loop
Infinite to 10,000 ohms leak
15 to 26 1/2 v d c

b. Operate hand generator of calling telephone.

c. Associated Line Signal shall go to steady "White" condition.

Night Alarm (NA) buzzer shall also sound when "NA" switch is in "AUD" position or panel lamps shall light as a visible night alarm if "NA" switch is in "VIS" position.

d. Insert "ANS" plug (rear) into calling party's jack (line Signal shall go Black) and the night alarm indication should go out. Operate the associated "TALK-List" key.

Operator and calling party should be able to talk ("Operator's Talk Bat" key in "ON" position.)

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4.7.1.6.1 Magneto line test (Contd)

e. Insert "CALL" (front) plug of same cord circuit into jack associated with called party's line. ("TALK-LIST" key operated.)

f. Ringing is accomplished by methods listed below:

(1) To ring called party:

Operate ring key in Telephone Circuit Operator's TA-220( )/P to "RING FWD" Position and turn crank of hand generator.

(2) To ring calling party, same as above but operate ring key to "RING BACK" position.

g. When calling party answers, both parties and operator should be able to talk. Restore talk key to normal, operator shall be disconnected.

h. Disconnect

(1) Operation of generator crank at magneto (calling) station should cause only the associated supervisory signal in Telephone Circuit, Cord TA-208( )/P to a steady "White." (The other signal should not flutter.)

(2) When plugs are restored to the cord well or the talk key operated to talk, the supervisory signals shall go to a steady "Black" and not blink.

The above conditions are to be met when the three supervisory signals of the same cord pack are black or white, on loops of 0-5000 ohms with infinity to 10,000 ohms leakage.

4.7.1.6.2 Common battery signaling, (CBS) line test:

a. On a CBS line of 0 to 1000 ohms loop and infinity to 10,000 ohms leak at switchboard voltages of 20 to 26.5 volts, a disconnect on a supervisory signal shall not come in as a flashing signal when encircling five supervisory signals are either in the white or black condition but shall follow the hookswitch. (The most severe condition is 0 ohms loop and 10,000 ohms leak at 26.5 volts.)

b. On a CBS line of 0 to 1000 ohms loop and 10,000 ohms leak at switchboard voltages of 20 to 25 volts with encircling five supervisory signals white, failure of a disconnect signal to come in until adjoining cord circuits are cleared will not be acceptable.

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4.7.1.6.2 Common battery signaling, (CBS) line test:

c. Supervisory signals in cord circuits connected to CBS lines of 0 to 1000 ohms loop and infinity to 10,000 ohms leak at switchboard voltages of 21 - 26.5 volts and 750 ohms loop at 20 volts with encircling five signals black, the signal shall operate (follow the hookswitch). The most severe condition is 1000 ohm loop and infinity leak at 21 volts.

d. On a CBS line with a 10,000 ohms leak, blinking or continuous steady white condition of a line signal if the calling party abandons a call before the operator answers, will not be acceptable. Line signals should operate to white on a 1000 loops and infinite to 10,000 ohms leak at 20 to 26.5 volts and should follow the hook switch.

e. Supervisory and line signals shall not "hang up" when the hook switch on a Telephone Set TA-312( )/PT is operated 10 time consecutively at the rate of one operation per second. This test shall be repeated three times within a period of three minutes.

f. Lifting of handset on calling telephone should cause associated line signal on switchboard to go "White". Night Alarm "NA" signal same as listed under Magneto test.

g. Completion of call handled in same manner as outlined under Magneto test.

h. With cords "up", replacing handset of station telephone should cause only the associated supervisory signal to go "White".

i. Restoration of plugs to cord well or operation of Talk key to Talk should cause supervisory signals to go "Black."

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4.7.1.6.3 Common battery signaling (CBS) trunk test: (See 3.6.1.6c)

a. Limits:

0 - 2000 ohms loop
Infinite - 25,000 ohms leak
20 -26.5 volts dc

Connect an SB-86( )/P Switchboard to a second SB-86( )/P Switchboard, each board having its own battery supply. The trunk connections should be polarized (connect tip to tip and ring to ring).

b. Insert "CALL" plug into trunk jack at originating switchboard and operate associated key of cord circuit to "TALK-LIST" position, and operate the "TRK-SIG" key.

c. The trunk signal at the terminating switchboard should go "White", when the "TRK-SIG" key at the originating switchboard is operated. A signal which goes to the white position, as above but does not follow the operation of the TRK-SIG key will be acceptable.

d. Plug an "ANS" cord into the trunk jack associated with the trunk signal at the terminating switchboard. The trunk signal should go "Black."

e. Restore talk key at originating switchboard, both cord supervisory signals should remain black.

f. Operation of the talk key and the "TRK-SIG" at either switchboard should not affect the steady black supervisory signals.

g. Either operator may disconnect. The cords can be pulled down with or without the "Talk-List" key operated. The first operator to disconnect should cause the supervisory signal of the other switchboard to go white. The line signal of the first switchboard should remain black (should not blink). Pulling down of the cords on the switchboard having a white supervisory signal associated with the trunk and restoration of the plug to the cord well will cause the supervisory signal to go a steady "Black" and should not disturb the black line signal on the originating switchboard if the Talk key has not been operated. There shall be no continuous blinking of the trunk signal at the first switchboard which disconnects when the "Talk-List" key at the distant switchboard is operated and restored or when the plug is removed from the jack. A momentary blink of the trunk signal under these conditions will be acceptable.

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4.7.1.6.3 Common battery signaling (CBS) trunk test: Contd.-

h. If the first switchboard operator reseizes (that is, replugs into the line that was disconnected before the distant operator pulls down) the white supervisory signal will remain white and when the distant operator pulls down the line signal will come in as a steady white.

i. Operation of the "TALK-LIST" key on a trunk connection (MAG, or CBSTK) shall not cause false operation of a line or supervisory signal on any other connection.

NOTE: Test above at 20 volts and 26.5 volts at both switchboards. The above requirements should be met when surrounding supervisory signals are white or black.

4.7.1.6.4 Civilian trunk connection.- (Line 29 or 30 only) With line selector switch in "IN" position and with a dummy plug in the trunk jack, across line 29 or 30.shall read approximately 50 ohms with CIVTRKS switch in ON position and open circuit with switch in OFF position.

4.7.1.6.5 Conference connection.-

a. Ring telephone of each party required for the conference - inform each party to stand by for a conference (both cords of a cord circuit can be used).

b. Connect cord circuits together by operating the talk key to "Conf.". All stations should be able to talk. The operator may monitor by operating the "Conf-Supv" key.

c. Cord supervision with key in "Conf" position as described in the preceding paragraphs should still be obtained regardless of type of circuits connected to.

d. Any number of cord circuits may be switched to the conference circuit.

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4.7.2 Power Supply PP-990( )/P (See 3.6.2).- Install ten each Batteries BA-200( )/U in the equipment battery compartment. Connect a VTVM having a minimum input impedance of 10 megohms and a calibrated accuracy of 0.2%, across the 24 volt output terminals. Connect an adjustable DC power supply having output voltage and current meters across the External Battery input terminals. Connect two 1000 ohm, 10 watt resistors across the Ringing Supply output terminals with a switch so that either 2000 or 1000 ohms may be selected. Connect an oscilloscope across the Ringing Supply whose vertical scale has been calibrated to measure the 20 Hz square wave voltage. An adjustable 20 Hz oscillator shall be available for connection to the horizontal terminals of the oscilloscope to obtain Lissajous figures.

4.7.2.1 Metering.- Operate the EXT-INT BATT switch to INT. Operate BATT CHECK switch to position 2. The voltmeter reading shall be within 1.0 volts of the VTVM reading which should be approximately 30 volts.

4.7.2.2 DC output voltage.- Operate BATT CHECK switch to position 1 and INT SWBD BATT switch to HI. The voltmeter shall read approximately 30 volts. Operate INT SWBD BATT switch to LOW. The voltmeter shall read approximately 24 volts. The voltage across terminals E601 and E602 should be approximately 6 volts. Operate EXT-INT BATT Switch to EXT. The voltmeter shall read the voltage of the external power supply in both positions 1 and 2 of the BATT CHECK switch. There shall be no voltage across terminals E601 and E602.

4.7.2.3 AC output voltage.- With power supplied by the external battery, effect closure between terminals E605 and E606 (ST VIB). Measure and display 20 Hz ringing-voltage for voltage, wave shape and frequency to meet the requirements of 3.6.2.3.

4.7.3 Insulation Resistance.- With input and output terminals of line packs, cord packs and the operators pack shorted, measure the insulation resistance between each circuit and chassis ground by Method 302 of MIL-STD-202, test condition A to meet the requirement of 3.6.3.

4.8 Service conditions tests.-

4.8.1 Temperature test (See 3.7.1).-

4.8.1.1 Low temperature.- The equipment shall be tested in accordance with Method 502, Procedure 1, of MIL-STD-810. The storage temperature shall be -80°F and the operating temperature shall be -40°F.

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4.8.1.2 High temperature.- The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 150°F.

4.8.2 Humidity (moisture-resistance) test.- The equipment shall be tested in accordance with Method 507, Procedure II of MIL-STD-810 (See 3.7.2).

4.8.3 Altitude test.- The equipment shall be tested in accordance with Method 500, Procedure I of MIL-STD-810 (See 3.7.3).

4.8.4 Leakage (immersion).- The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I of MIL-STD-810 (See 3.7.4).

4.8.5 Fungus test.- The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required (See 3.7.5).

4.8.6 Vibration and Bounce.-

4.8.6.1 Vibration (Resonance search).- The equipment shall be tested in accordance with Method 514, Procedure XI, Part I, of MIL-STD-810 to determine conformance to 3.7.6.1.

4.8.6.2 Bounce test.- The equipment, installed in the Cover Assembly, shall be tested in accordance with Method 514, Procedure XI, Part 2, of MIL-STD-810. At the completion of the bounce test, the equipment shall be tested for compliance with 3.7.6.2.

4.8.7 Shock test; bench handling.- The equipment shall be tested in accordance with Method 516, Procedure V of MIL-STD-810 (See 3.7.7)

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4.9 Inspection for interchangeability.- The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.8. When listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of covers, and insertable subassemblies when such dimensions affect mating of parts.

b. Internal dimensions of cavities in assemblies, when such dimensions affect acceptability of insertable subassemblies.

c. Location of fasteners on separable parts of subassemblies which must mate together, such as covers and mounting.

d. Locations of connectors, fasteners and mountings which receive mating parts of plug-in assemblies and major components.

e. Locations of connectors, fasteners, and mountings on plug-in assemblies.

4.10 Air seal test (See 3.11).- The equipment shall be opened and closed again in such a manner as to break and remake the seal. Immediately thereafter, the equipment, as field transported, shall be subjected to a vacuum of one pound per square inch (one pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment enclosure. The vacuum shall be valved off and interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

4.11 Visual and mechanical inspection.- The units shall be examined for conformance to the construction drawings, to the workmanship requirements (see 3.14) and for the defects listed in MIL-STD-252.

4.12 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to requirements of Section 5. If the rough handling test is specified in the bid request and contract, the equipment shall meet the requirement of 3.6.1 and 3.6.2 after this test.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C, as specified. (See 6.2)

5.1.1 Level A.

5.1.1.1 Cleaning.- Each Switchboard, Telephone, Manual SB-86( )/P shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Each Switchboard,Telephone Manual SB-86( )/P shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Spares.- Place the spares within the case and secure to prevent damage.

5.1.1.5 Each Switchboard, Telephone Manual SB-86( )/P, shall be packaged in accordance with items 1 through 5 of Figure 1, Table VI.

5.1.2 Level C. Each Switchboard, Telephone Manual SB-86( )/P, shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing.- Packing shall be level A, B or C as specified (See 6.2.b).

5.2.1 Level A.-

5.2.1.1 Switchboard, Telephone Manual SB-86( )/P, shall be packed in accordance with Item 6 and 7, of Figure 1, Table VI.

5.2.2 Level B.- Switchboard shall be packed as specified in 5.2.1.1.

5.2.3 Level C. Switchboard, Telephone Manual SB-86( )/P, shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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BILL OF MATERIAL
TABLE VI

REF No	ITEM Nomenclature	No. Reqd	Method MIL-P-116	SIZE - INCHES L	W	D	Spec.No.	REQUIREMENT T	G	C	S	Flute	PSI
1	Technical Manual	2	IC-1	10	8		MIL-B-117	11		D			
2	Switchboard SB-86()/P	1	III	24	18-1/2	23							
	Fuse												
	Block												
	Lamp												
	Fuse												
	Lamp												
	Plate												
	Hand Headset												
	Cable												
	Cable												
	Case												
	Place Spares in Case and Secure to prevent damage.												
3	Top and Bottom Pad Fiberboard	2		25-1/2	19-5/8	3/8	PPP-F-320	CF	W5	Variety		SW	
4	Corner Pads Fiberboard	4		23-1/2	8	3/8	PPP-F-320	CF	W5	Variety		SW	
5	Box Fiberboard	1		25-1/2	19-3/4	24-1/2	PPP B 636	1	5	2	RSC	C	
	Tape Pressure Sensitive	2		32	3								
		4		26	3		PPP-T-76						
		1		22	3		PPP-B-621			2			
6	Box Nailed Wood or Box Cleated Plywood	1		26-1/4	20-1/2	25-1/4	PPP-B-601	111		2	A		
7	Strapping Steel	2		104	5/8	.020	QQ-S-781	1	2	B			

LEGEND
T - Type
G - Grade
S - Style
C - Class

PACKED WEIGHT 240 Lbs.
PACKED VOLUME 10.0 Cu. Ft.

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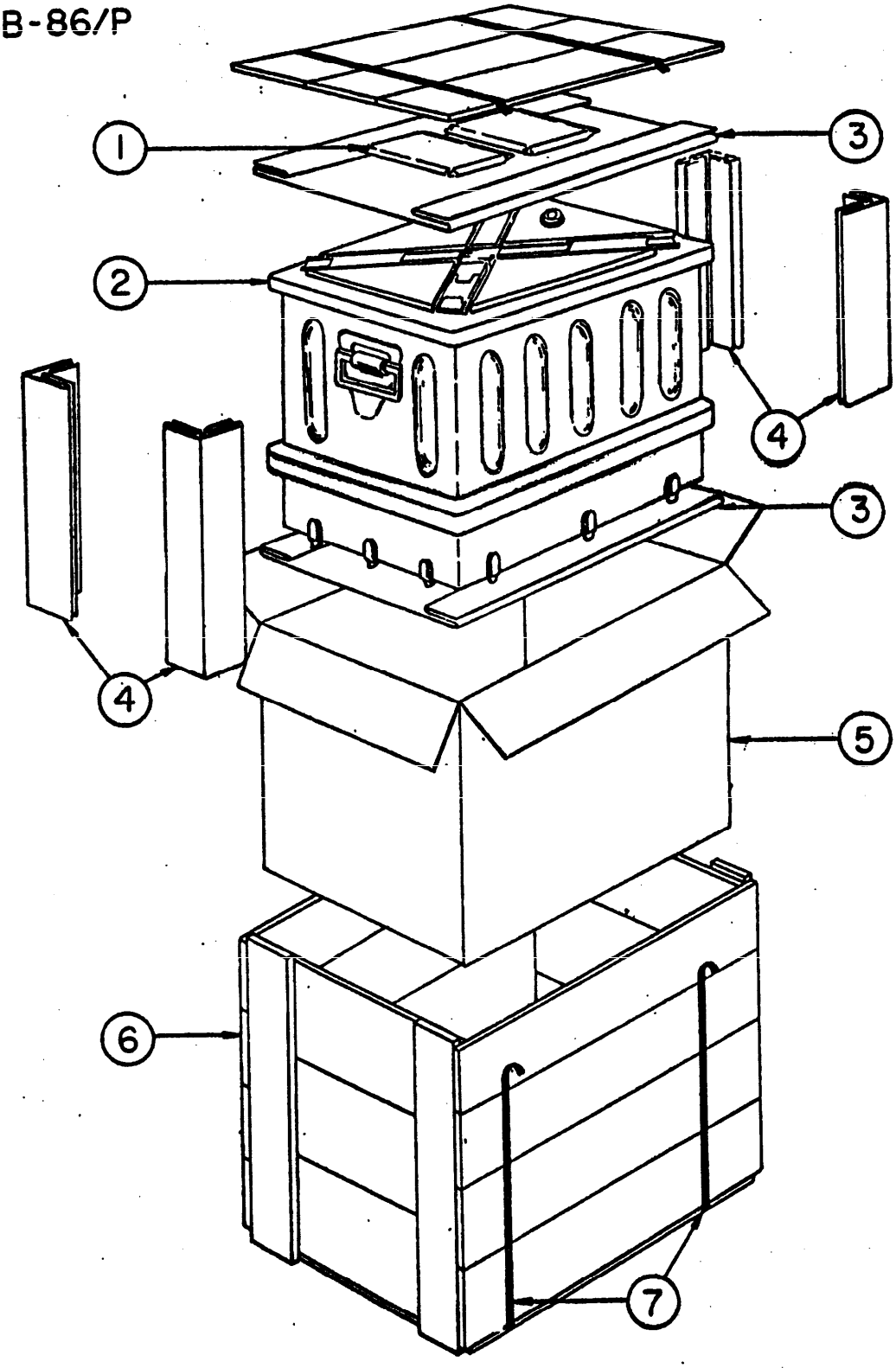


FIGURE 1

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6. NOTES

6.1 Intended use.- Switchboard, Telephone, Manual SB-86( )/P is a transportable switchboard for field use, capable of serving 30 lines. By using two jack field section, 60 lines may be served.

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, date of this specification and any amendment thereto.

b. Level of packaging and packing required for shipment (Level A, B or C).

c. When rough handling and functional tests are required.

d. Samples required.

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required.

h. Quantity of running spare parts.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: "SB-86W/P". The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian
Army - EL

Preparing Activity
Army - EL

Project Number 5805-
A166

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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